

Cliff Diving: Exploring Reward Surfaces in Reinforcement Learning Environments

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2022/12/18

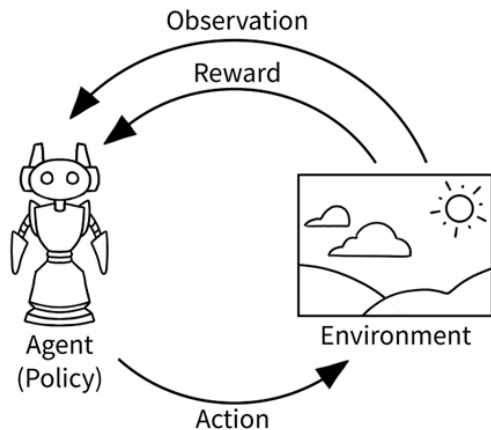
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2. Background
3. Environment selection
4. Initial explorations of reward surfaces
5. Discovering cliffs in the gradient direction
6. Cliffs impact policy gradient training
7. Discussion

1. Introduction

강화 학습 과정은 안정적으로 진행되지 않음

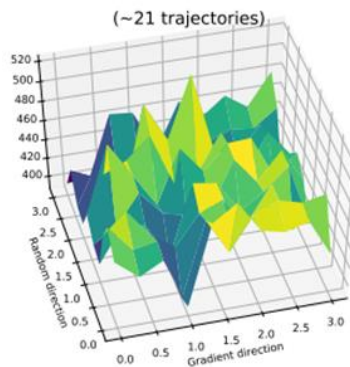


Source: Multi-Agent Deep Reinforcement Learning in 13 Lines of Code Using PettingZoo

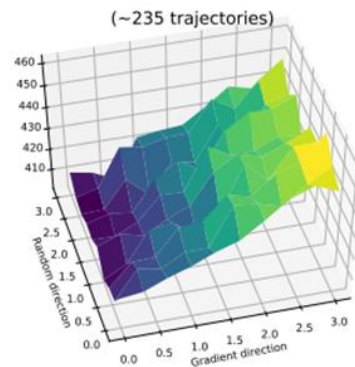
강화 학습의 목적

- 에이전트의 누적 보상 극대화를 위해 가장 최적화된 학습 정책(policy) 찾는 것이 목표
- 일반적인 심층 학습에 비해 학습 과정이 안정적이지 못함

2,000 state-action pairs



20,000 state-action pairs

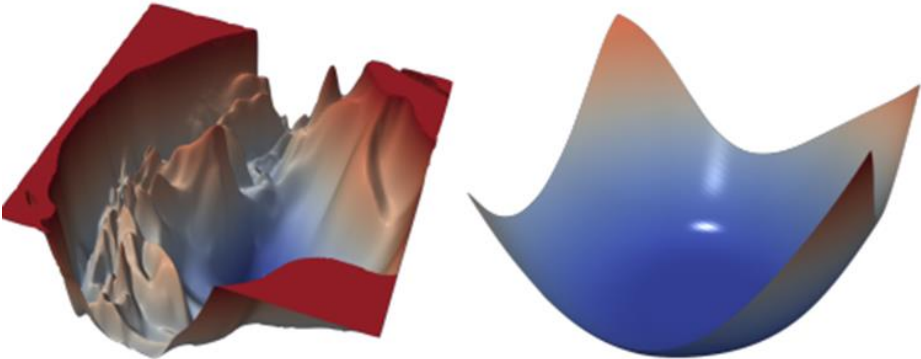


A closer look at deep policy gradients [ICLR2020]

보상 표면 (reward surface)

- 강화 학습의 목적에 따른 각 보상 값을 고차원 표면으로 시각화 (ilyas et al., 2020)

시각화를 통해 다양한 변수의 영향을 손쉽게 확인



(a) ResNet-110, no skip connections
Figure 4: The loss surfaces of ResNet-110-noshort and DenseNet for CIFAR-10.

Visualizing the Loss Landscape of Neural Nets [NIPS 2018]

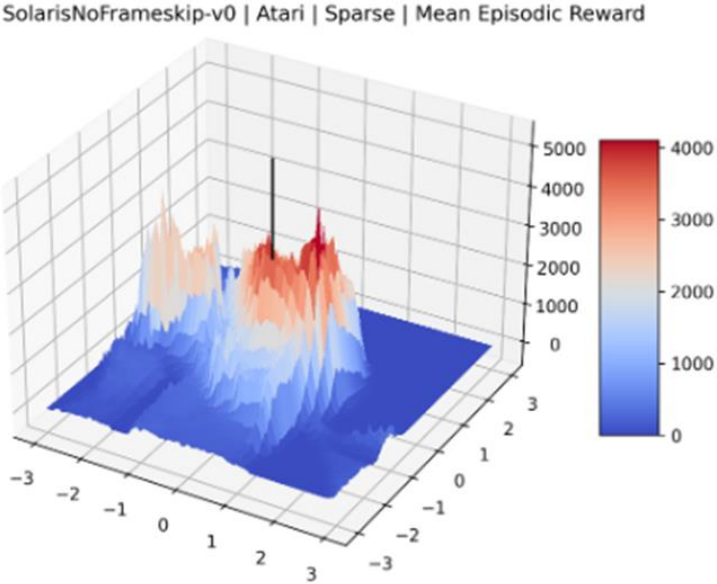


Figure 1. Reward surface for the Solaris Atari environment.

Cliff diving [ICML 2022]

2. Background

Loss landscapes 기법을 강화 학습에 적용

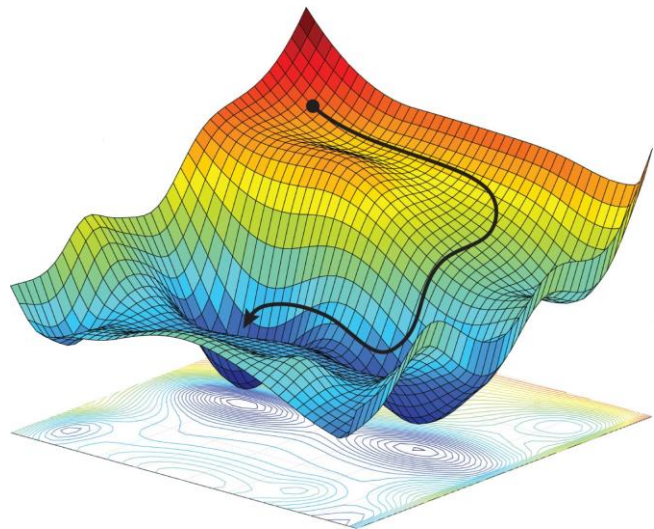
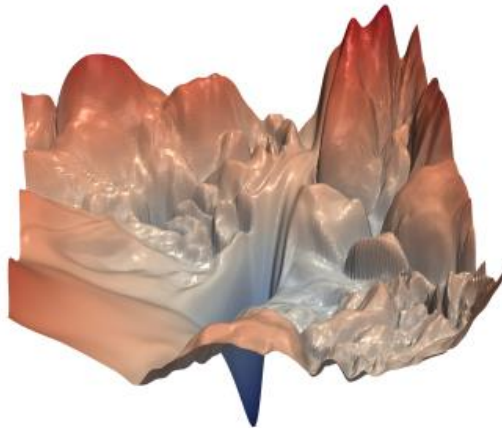
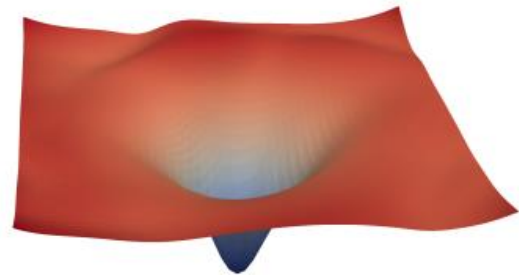


Image source: Has artificial intelligence become alchemy? [Science]



(a) without skip connections

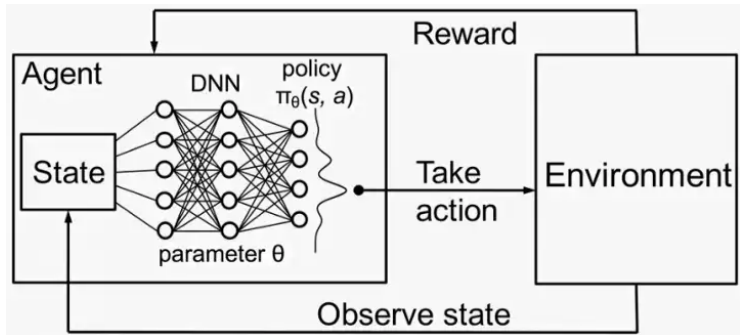


(b) with skip connections

Visualizing the Loss Landscape of Neural Nets [NIPS 2018]

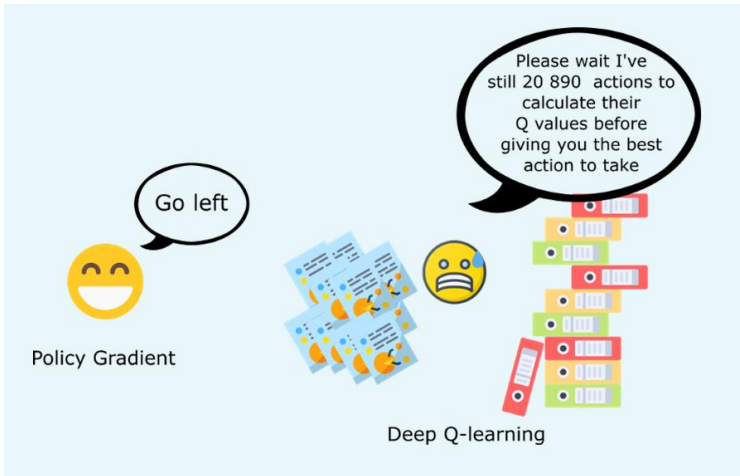
Reinforcement learning & policy gradient

심층강화학습 (Deep reinforcement learning)



Deep Reinforcement Learning: Value Functions, DQN, Actor-Critic method, Back-propagation through stochastic functions [Medium]

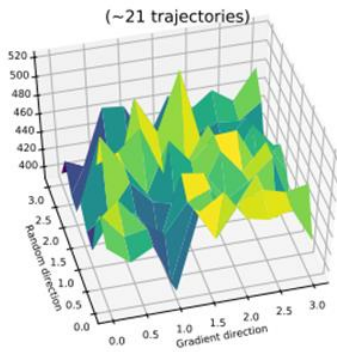
정책 경사 기기법(Policy gradient methods)



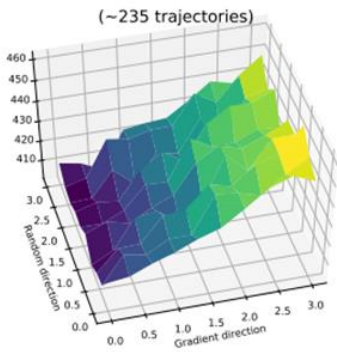
An introduction to Policy Gradients with Cartpole and Doom

Previous works on reward surfaces

2,000 state-action pairs



20,000 state-action pairs



A closer look at deep policy gradients [ICLR2020]

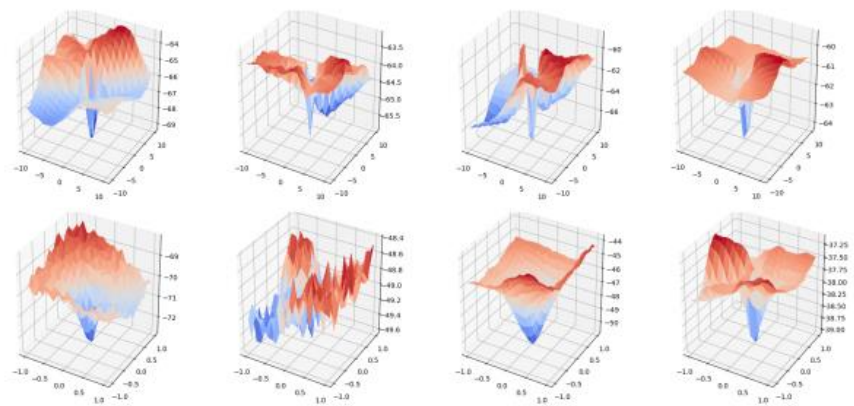


Figure 3: Actor loss functions of SAC trained on Walker2d, Hopper, Ant, and HalfCheetah. Stochastic policy on the upper row, deterministic policy on the lower row.

Visualizing the loss landscape of actor critic methods with applications in inventory optimization [arXiv 2020]

Proximal policy optimization (PPO)

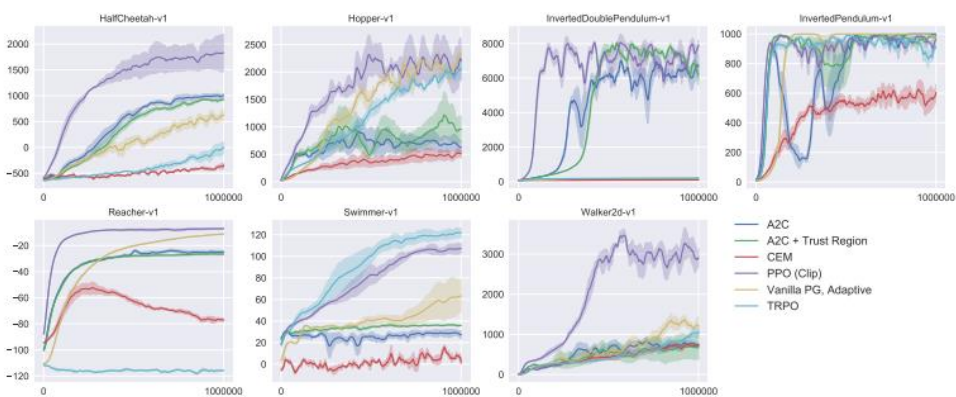
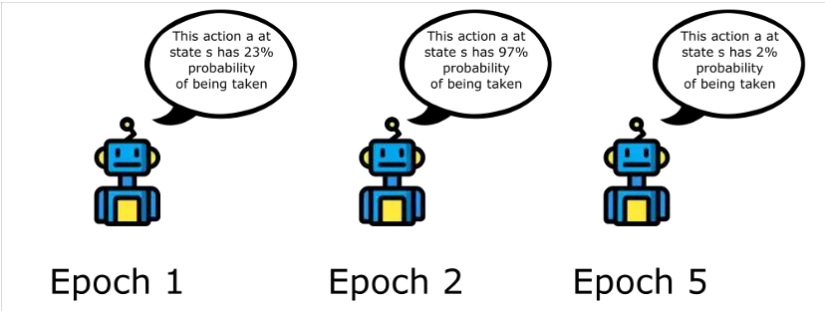


Figure 3: Comparison of several algorithms on several MuJoCo environments, training for one million timesteps.

Image credit: Proximal Policy Optimization (PPO) with Sonic the Hedgehog 2 and 3, towardsdatascience.com

Proximal Policy Optimization Algorithms [arXiv, 2017]

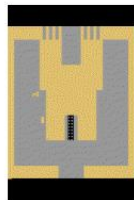
Q & A

3. Environment selection

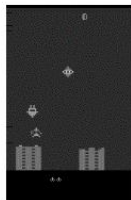
Environment selection

Atari

A set of Atari 2600 environment simulated through Stella and the Arcade Learning Environment.



Adventure



Air Raid

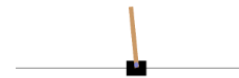


Alien

Classic Control



Acrobot

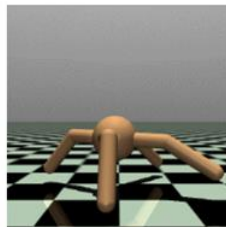


Cart Pole

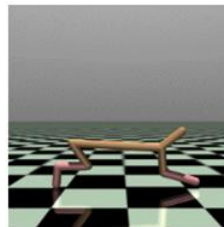


Mountain Car
Continuous

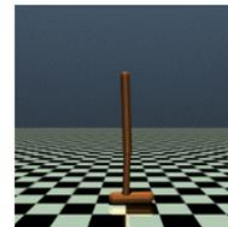
MuJoCo



Ant

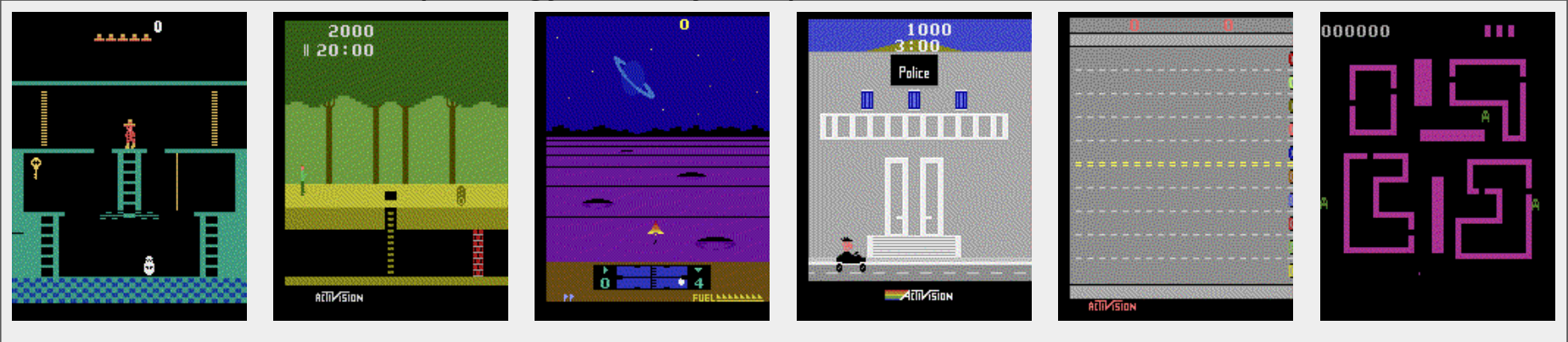


Half Cheetah

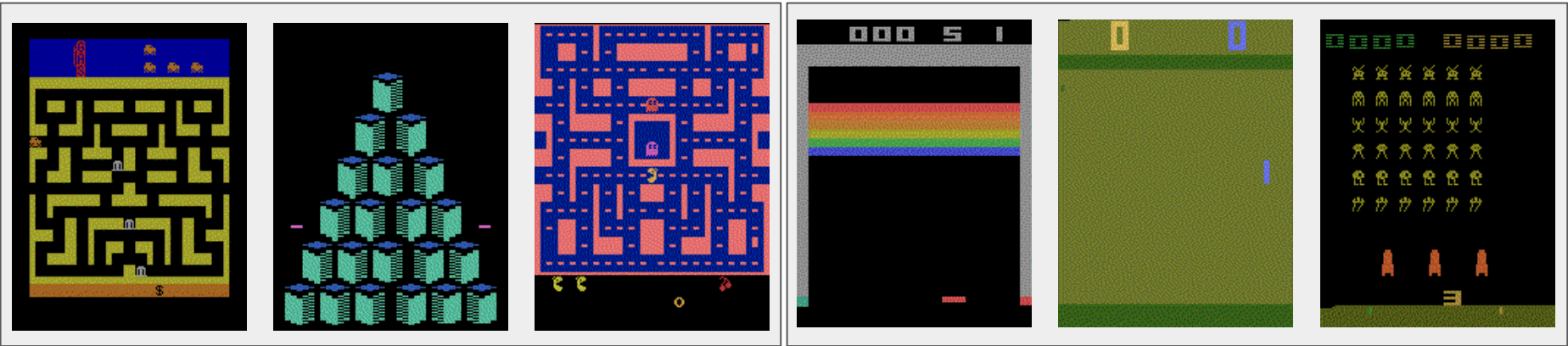


Hopper

Environment selection (www.gymlibrary.dev)



Sparse reward environment



dense reward environment

Easy exploration environment

Top (left to right): Montezuma's Revenge, Pitfall!, Solaris, Private Eye, Freeway, Venture
Bottom (left to right): Bank Heist, QBert, Ms. Pac-Man, Breakout, Pong and Space Invaders

4. Initial explorations of reward surfaces

Results

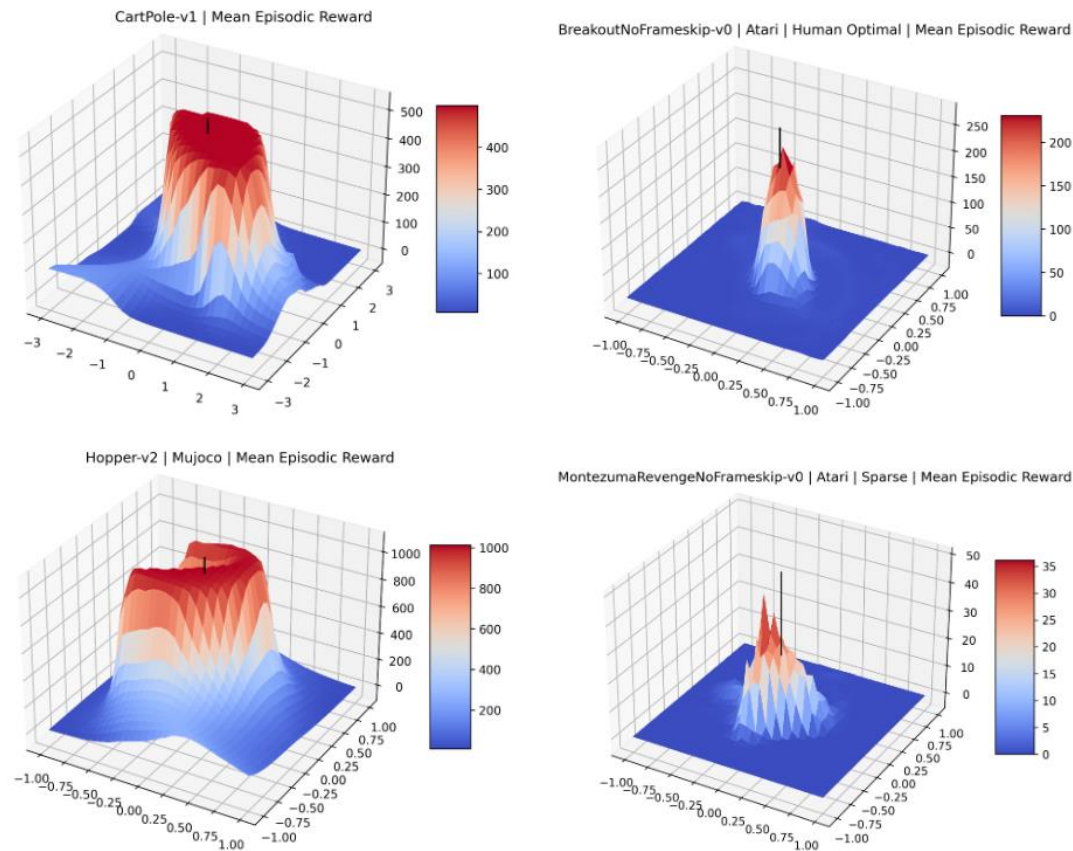


Figure 2. Reward surfaces for CartPole-v1, Breakout, Hopper-v2, Montezuma’s Revenge

Results

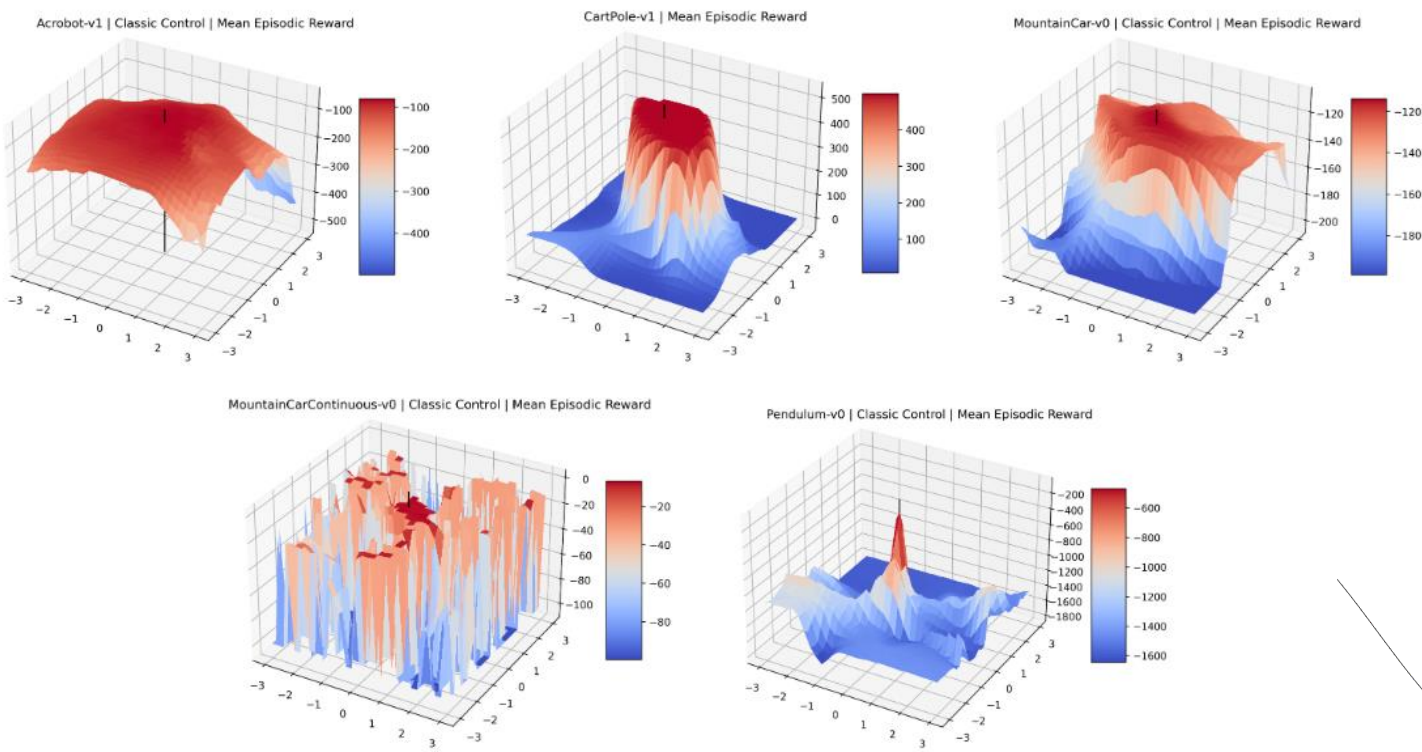
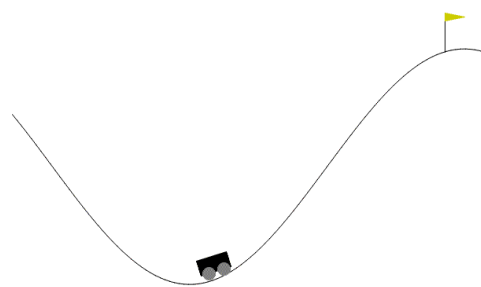


Figure 5. Reward surfaces for 5 Classic Control environments.



Mountain Car Continuous [Gym]

Results

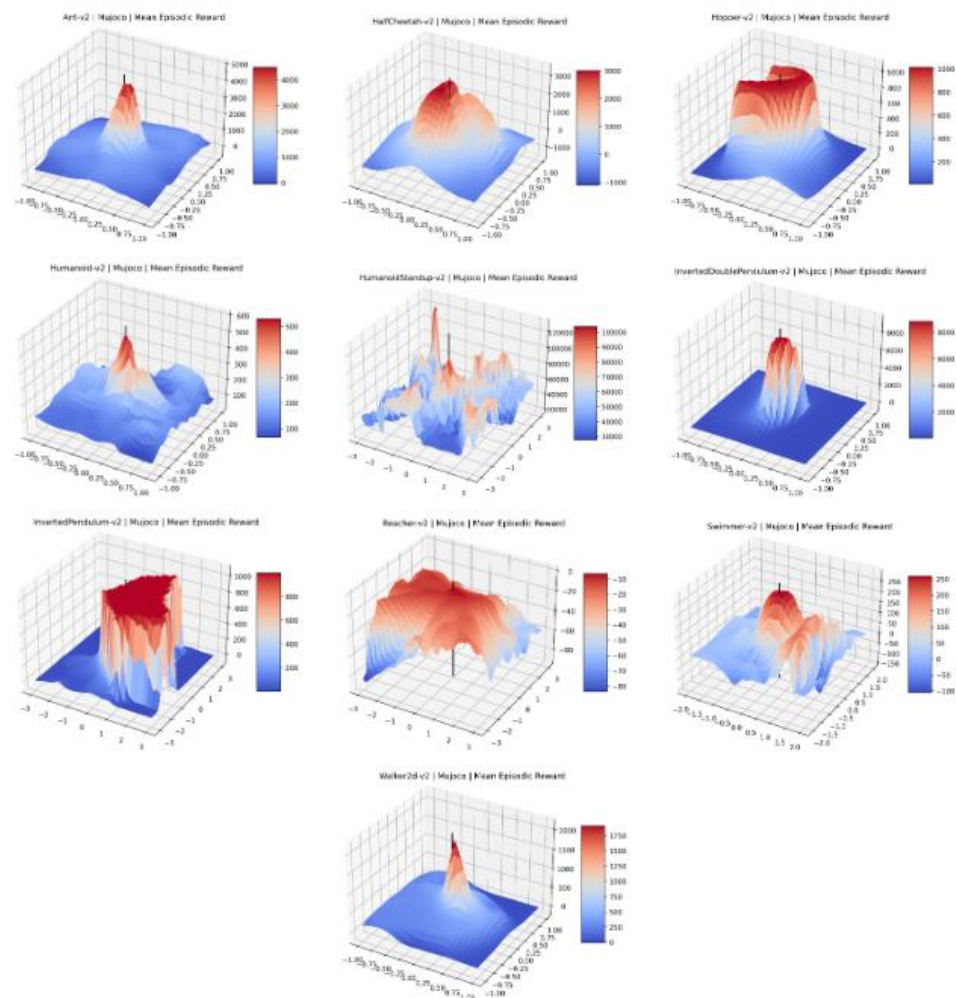


Figure 6. Reward surfaces for 10 MuJoCo environments.

Results

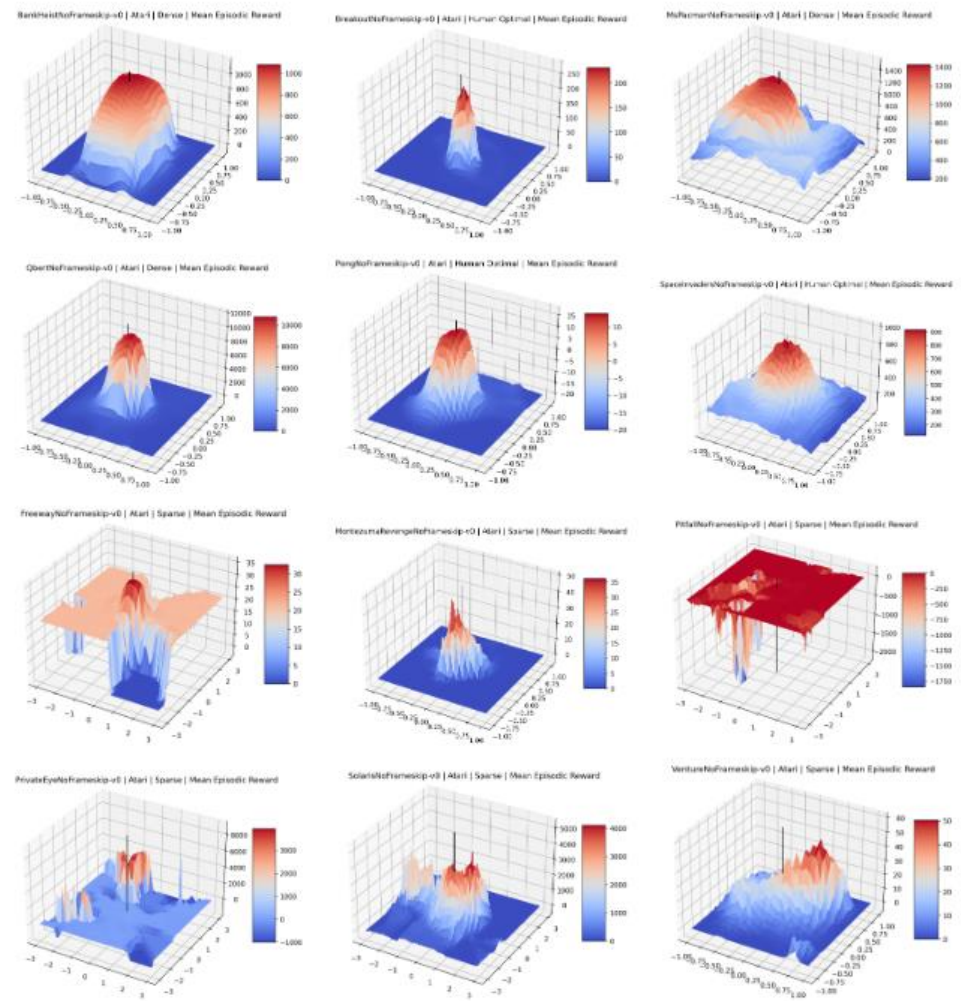


Figure 7. Reward surfaces for 12 Atari environments.

5. Discovering cliffs in the gradient direction

Gradient detections vs. filter-normalized random directions

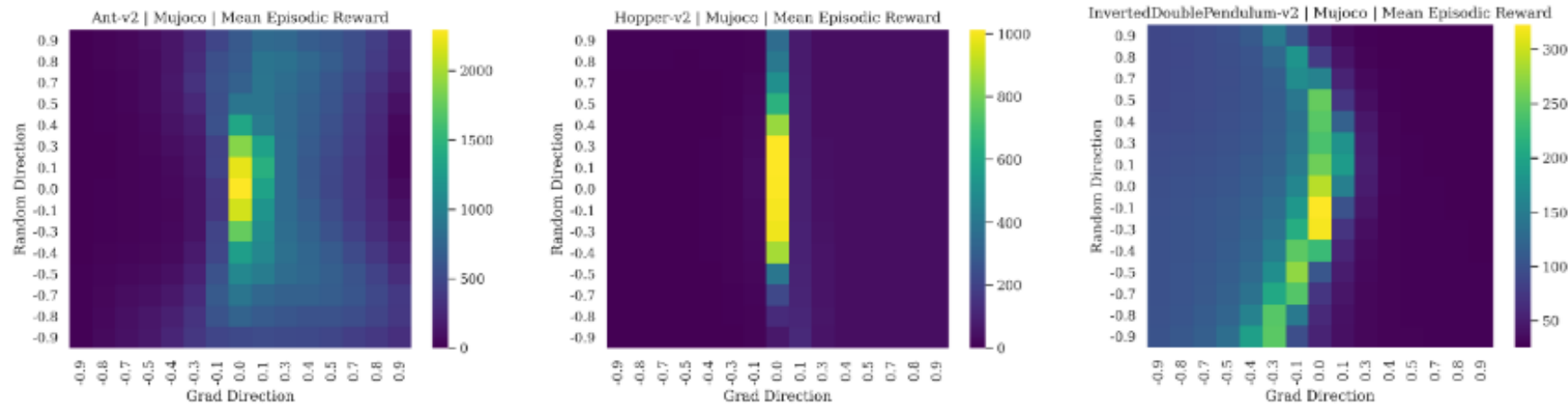


Figure 3. Gradient heat maps for Ant, Hopper, and InvertedDoublePendulum. The heat map for Hopper and InvertedDoublePendulum shows a cliff-like gradient direction which falls off sharply compared to the filter-normalized direction.

Visualizing rewards in the gradient direction during training

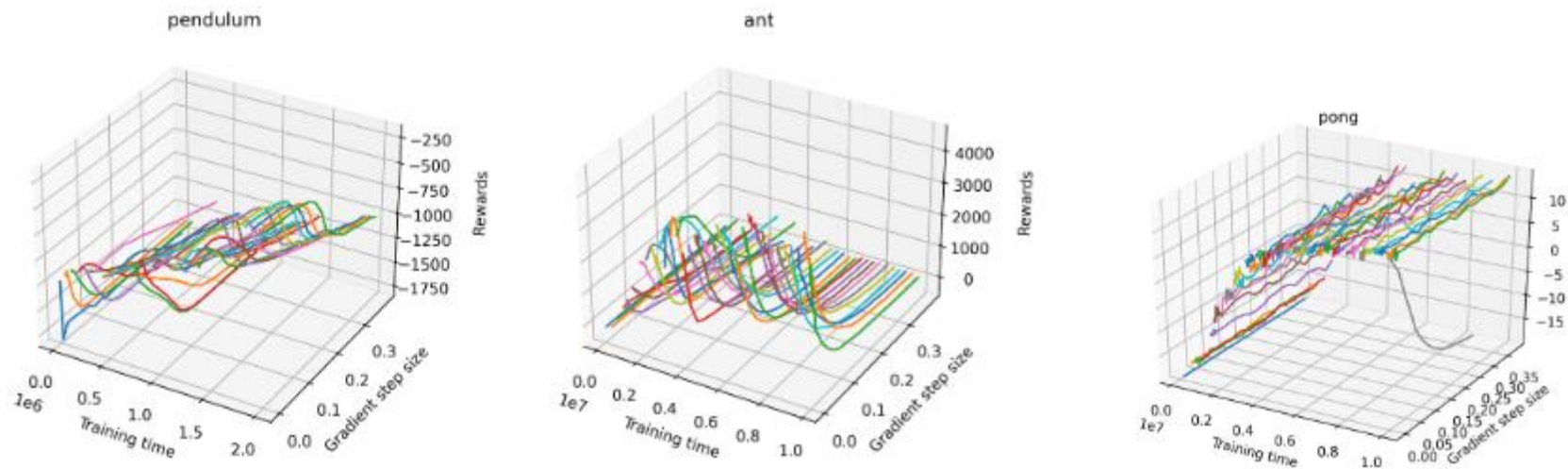


Figure 4. Gradient line search plots for Pendulum, Ant, and Pong. The line plot for Ant shows several checkpoints that exhibit cliff-like properties.

Q & A

6. Cliffs impact Policy Gradient Training

Methodology + Results

			Cliff	Non-Cliff
N steps = 128	LR = 0.000001	PPO	0.03%	0.03%
		A2C	-0.3%	0.2%
	LR = 0.0001	PPO	-0.01%	-0.03%
		A2C	-4%	2.3%
N steps = 2048	LR = 0.01	PPO	0.0%	0.4%
		A2C	-0.3%	2.0%
	LR = 0.5	PPO	0.0%	0.0%
		A2C	-8.5%	1.6%
N steps = 2048	LR = 0.000001	PPO	-0.1%	-0.4%
		A2C	-0.5%	0.2%
	LR = 0.0001	PPO	-0.2%	-0.1%
		A2C	-7.0%	4.4%
N steps = 2048	LR = 0.01	PPO	0.1%	0.1%
		A2C	-3.9%	2.9%
	LR = 0.5	PPO	0.0%	0.3%
		A2C	-3.6%	-0.3%

Table 1. Table of A2C and PPO's average percent change in reward after taking a few gradient steps on cliff and non-cliff checkpoints for various sets of hyperparameters. These results are averaged among 10 trials each evaluated for 1000 episodes.

7. Discussion

Discussions

- CV 분야 학습에서 Loss landscape 은 디버깅용으로 유용
- 강화학습에서도 reward surface 유사 용도 활용 기대
- 급격한 절벽 현상이 학습 진행에 어떤 결과를 주는지 시연하였음
- PPO와 A2C 모델 학습 능력 비교 시 PPO가 왜 더 효과적인지 확인
- 향후 multiagent 학습에서는 본 시각화 어떤 양상을 보일지 기대

Q & A